UQBR Aquaria SOP 3 –Feeding Protocols of Zebrafish

REQUIREMENT:

1. To ensure that the requirements and regulations as set out by the following are met as far as practicable:
   - AEU UQ
   - The Code
   - OGTR
   - Department of Agriculture and Fisheries (DAF)
   - QLD Workplace Health and Safety, and
   - UQ OH&S
2. To standardise practice for all UQBR staff and researchers within UQBR facilities.
3. Annual review is required to maintain best practice and usability of this SOP.

RESPONSIBILITY:

It is the responsibility of the individual performing animal handling procedures and techniques to ensure they have been assessed as competent.

Please Note:
This UQ Biological Resources (UQBR) SOP expands upon UQ Animal Ethics Unit SOPs. This document outlines the procedures followed by UQBR and should not be referenced in Animal Ethics Applications.

No changes or deviations from this SOP are to occur unless the Director of UQBR gives prior authorisation.

NB: The use of (*) indicates this statement is dependent on the facility procedures
NB: The use of (**) indicates this statement is dependent on AEC Approvals

OBJECTIVE:

Ensuring that zebrafish are raised on the appropriate density and feeds for their current stage of ontogeny. They will grow evenly and reach sexual maturity within current best practice timeframes (>90days).

I. EQUIPMENT

Equipment Items

- Manual food clickers (Figure 1)
- Transfer pipettes

Consumables

- Rotifers (Type-L marine (*Brachionus plicatilus*))
- Fry Food Mix
- Juvenile Food Mix
- Adult Food Mix

Administration

- Manual
- Automatic (Triton)- *Refer to SOP18 Tritone operation*

Figure 1 – Manual food Clickers

II. PREPARATION *(Refer to Aquaria SOP 4 Production and Maintenance of Zebrafish Diets)*

Larvae Diet (Polyculture)

1. Harvested live Rotifers.
2. Create Polyculture.

Fry Food Mix.

1. Harvested live Rotifers.
2. Dry Food Preparation (O.range start + Spirulina powder)

Juvenile Food Mix.

1. Dry Food Preparation (O.range Wean + Fry Diet)

Adults Food Mix.

1. Dry Food Preparation (3kg O.range Wean+ 3kg NRD3/5 + 300gr Spirulina powder)

III. PROCEDURE (Refer to table 1 & Figure 1)

It is important to not overfeed tanks and to consider that Zebra fish are an agastric (stomach less) species. Adhering to the given feeding plan, portions and times will provide higher food conversion ratios and cleaner environments leading to healthier fish. Feed rate is roughly 5% the animal's bodyweight daily.

Larvae (~5-12dpf)
Feed: Polyculture (Live rotifer and algae mix*)
Location label: "Week 1 static tank"

1. Once the swim-bladder has inflated (~-5 days old):
   i. Place larvae into polyculture.

2. Visually assess density of Rotifers within the tank daily.
   a. If Density of Rotifers is low, top-up tank with **5ml** of adult Rotifers.
   b. If Density is sufficient re-assess the following day.

*Note: Rotifers alone offer no nutritional value to the fish, they are simply a means of bio encapsulating the nutrients we want delivered to the fish. As such, always ensure that the culture has been harvested and allowed to gut load on the algae mixture **at least 1 hour** before being used as a feed.

Fry (~12-19dpf)
Feed: Fry mix + Harvested rotifers
Location label: "Week 2 -3 on fry side of the nursery" slow drip.

1. Ensure fry are on a slow drip flow on the system (re-circulating system).

2. Feed Rotifers once per day
   a. Fed 5ml of rotifers Use a transfer pipette to top tank with rotifers as required.

   Note: Rotifers do not survive in the re-circulating system.
   Rotifers provided may be any age.

3. Feed fry mix two times per day. (8am and 4pm)
   a. Use correct size food clicker.
   b. On weekends Fry are fed once per day.

Small Juvenile (~20-25dpf)
Feed: Juvenile food mix.
Location label: "Week 3 on Juvenile side of the nursery".

1. Ensure juveniles have been graded and placed in larger tanks.

2. Feed Juvenile food mix three time per day
   a. If rotifers have been provided, feed the dry food towards end of shift to allow time to digest and prevent food spoiling and fouling of the tank.
   b. On weekends Juveniles are fed once per day.

   Note: Extra Rotifers may be fed if they are available.

Large Juvenile (~26-33dpf)
Feed: Juvenile mix + Adult mix.
Location label: "Weeks 4 - 5"

1. Ensure juveniles have been graded and placed in larger tanks.

2. Feed Adult feed and if necessary Juvenile feed three time per day.
   a. If rotifers have been provided, feed the dry food towards end of shift to allow time to digest and prevent food spoiling and fouling of the tank.
   b. On weekends Juveniles are fed once per day.

Note: Extra Rotifers may be fed if they are available.

Sub Adult (33-63 dpf)
Feed: Adult Feed mix
Location label: "Week 5-9"

1. Feed adult feed three times per day.

Adults (+63dpf, depending on facility)
Feed: Adult mix

1. **Feed adult food mix once per day in Seddon between 11-1pm**

Tritone Feeding – Fry, Juvenile and Adults

1. Tritone robots feeds dry food, at pre-set times based on the barcodes that distinguishes each tank by age group and stocking density.
2. The dose provided by the robot is based on the density range, this is determined by the feeder/label code. Each robot will feed the tank accordingly.
3. Ensure each feed bottles have the correct diet and is placed correctly (Refer to SOP 18).

IV. APPENDIX

Developmental Stage	Avg. body length	dpf	Diet	Delivery Method	Feeding Frequency	Feeding Times
Larvae	~3mm	~5-12	Larvae	Ad libitum	Grazing	Continual
Fry	~5mm	~12-19	Fry	Hand Fed	3x / day	8am, 12pm, 5pm
Small Juvenile	~10-15mm	~20-26	Juvenile	Hand Fed (T5 Bottle 4)	3x / day	8am, 12pm, 5pm
Large Juvenile	~10-20mm	~26-35	Adult + Juvenile	Hand Fed (Tritone)	3x / day	8am, 12pm, 5pm
Sub Adult	~15-25mm	~33-63	Adult	Hand Fed (Tritone)	3x / day	8am, 12pm, 5pm
Adult	25mm	63+	Adult	Hand Fed Tritone	1x / day 3x / day	12pm 8am, 12, 5pm

Table 1- Zebrafish Rearing Protocols

V. CONSIDERATIONS

1. Adverse events should be referred to UQBR SOP 22 UQBR Veterinary Care Protocol.
2. The goal is to feed 5% body weight per day to ensure maintenance diet requirements are met.
3. Food clickers manually dispense dry food and provide different quantities, the density of the tank guides the clicker and number of serves to be used.
4. Rotifers alone offer no nutritional value to the fish, they are simply a means of bio encapsulating the nutrients we want delivered to the fish. As such, always ensure that the culture has been harvested and allowed to gut load on the algae mixture **at least 1 hour** before being used as a feed.

VI. SAFETY

1. PPE use is essential when completing this task.
2. All accidents, injury or near misses are to be reported immediately to the Facility Manager and recorded on a UQ OHS Incident Report Form

VII. REFERENCES

1. Australian code for the care and use of animals for scientific purposes (8th Edition, NHMRC 2013): https://www.nhmrc.gov.au/guidelines/publications/ea28
2. Department of Agriculture and Fisheries (DAF): http://www.daf.qld.gov.au/
3. Guidelines to promote the wellbeing of animals used for scientific purposes (NHMRC, 2008): https://www.nhmrc.gov.au/_files_nhmrc/publications/attachments/ea18.pdf
4. OGTR PC2 work requirements and regulations: http://www.ogtr.gov.au
5. QLD WH&S Act 2011: https://www.worksafe.qld.gov.au/laws-and-compliance/workplace-health-and-safety-laws/laws-and-legislation/work-health-and-safety-act-2011
6. UQ Animal Ethics Unit SOPs: http://www.uq.edu.au/research/integrity-compliance/standard-operating-procedures-sops UQ OHS Unit: http://www.uq.edu.au/ohs/
7. UQ OHS Incident Report Form: http://www.uq.edu.au/ohs/index.html?page=141331
8. UQBR SOPs: V:UQBR/SOPs/Common/UQBR SOPs and http://biological-resources.uq.edu.au/secure/uqbr-sops